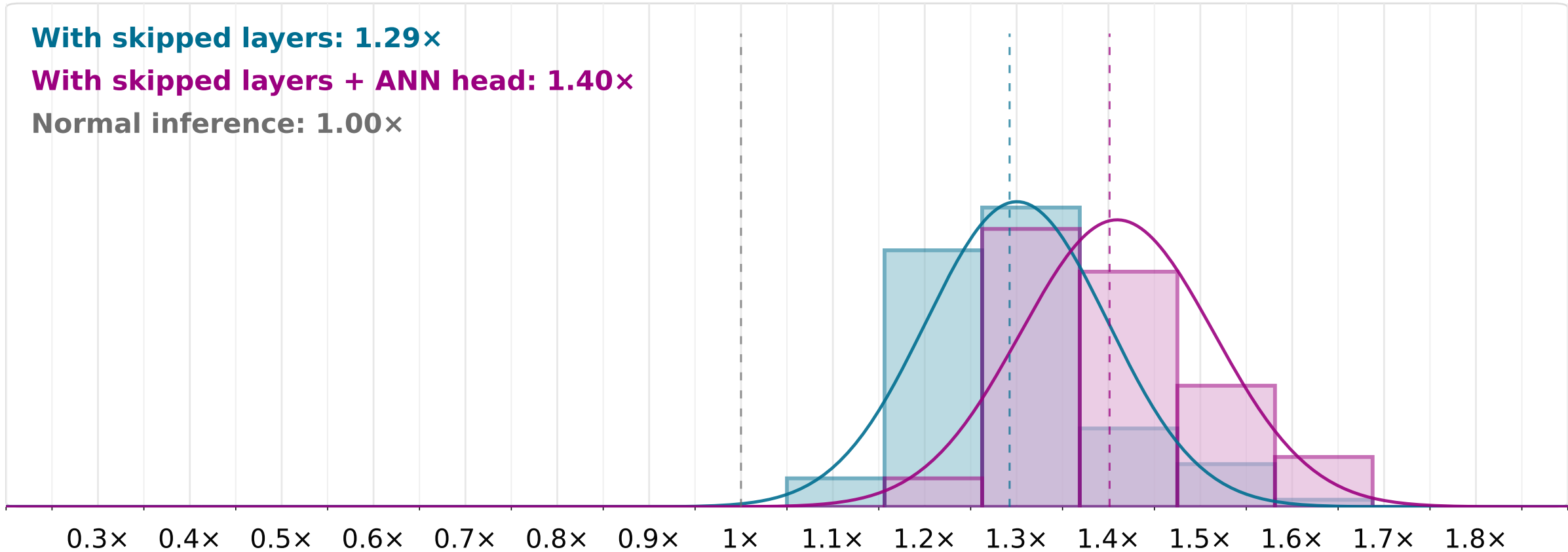


Self-speculation per-token speedup

Qwen3 4B Instruct 2507



Setup

Prompt set = **python diverse**
Gap = **(1, 1)**
Block size = **2**
Layers = **36**
LM vocab = **151,936**
LM params = **4.02B**
Head params = **389.0M**
Head portion = **9.67%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **100**

Results

Peak mem normal = **7.62 GiB**
Mean speedup (w/o ANNH) = **1.29×**
Acceptance rate (w/o ANNH) = **38.6%**
Exact match (w/o ANNH) = **53.0%**
Peak mem self (w/o ANNH) = **7.62 GiB**
Mean speedup (w ANNH) = **1.40×**
Acceptance rate (w ANNH) = **37.6%**
Exact match (w ANNH) = **53.0%**
Peak mem self (w ANNH) = **7.67 GiB**
Tokens gen = **s 19,836 / n 19,853**

Profile

Drafter split (w/o ANNH) = **B 42.7% / H 50.4% / O 6.9%**
Verifier/normal (w/o ANNH) = **1.07×**
Drafter/normal (w/o ANNH) = **15.9%**
Drafter split (w ANNH) = **B 75.8% / H 12.6% / O 11.6%**
Verifier/normal (w ANNH) = **1.07×**
Drafter/normal (w ANNH) = **8.8%**
ANNH accuracy = **97.5%**
ANNH head speedup = **7.13×**
ANNH index = **9496 clusters; top-k 100**